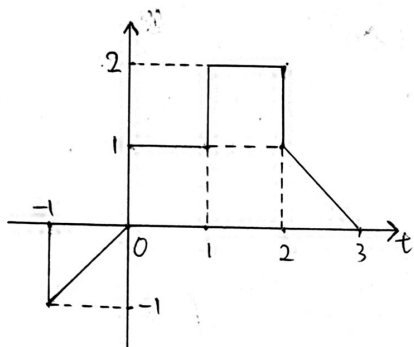


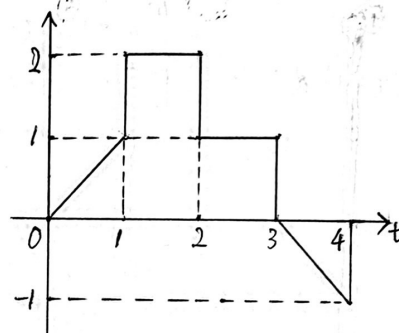


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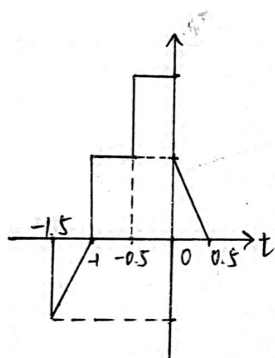
1.21 (a) $x(t-1)$



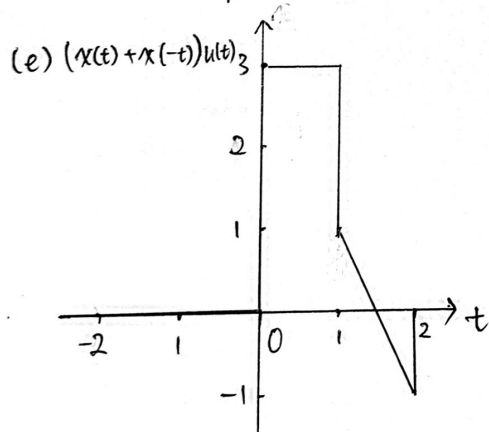
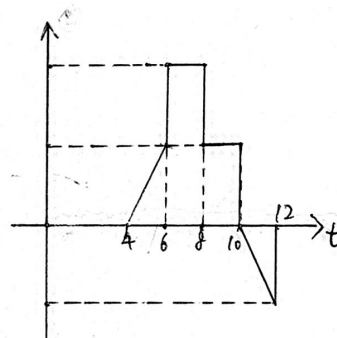
(b) $x(2-t)$



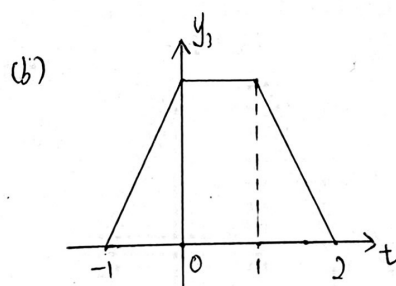
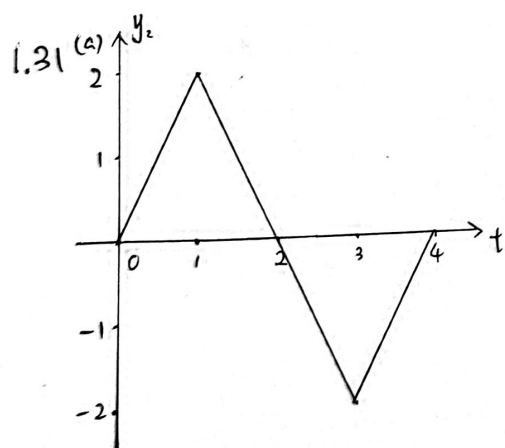
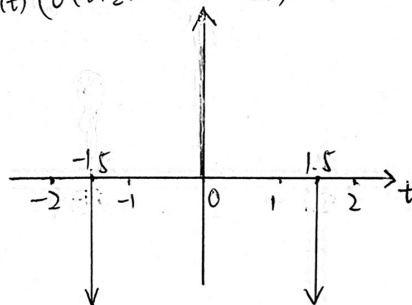
(c) $x(2t+1)$



(d) $x(4-\frac{t}{2})$



(f) $x(t) (\delta(t+\frac{3}{2}) - \delta(t-\frac{3}{2}))$



1.41 (a) 當 $g[n]=1$ 時, $y[n]=2x[n]$ (LTI).

(b) 當 $g[n]=n$ 時, $y[n]=n(n-1)x[n]$, 輸出結果與輸入次數有關

(c) 當 $g[n]=1+(-1)^n$ 時, $y[n]=2x[n]$ (LTI).

1.39 $\lim_{\Delta \rightarrow 0} u_{\Delta}(t) \delta(t) = \lim_{\Delta \rightarrow 0} u_{\Delta}(0) \delta(t) = 0$

$$g(t) = \int_{-\infty}^{+\infty} u(\tau) \delta(t-\tau) d\tau = \int_{-\infty}^{+\infty} u(\tau) \delta(\tau-t) d\tau = \int_t^{+\infty} u(\tau) \delta(\tau-t) d\tau = 0, t < 0$$
$$= \int_t^{+\infty} \delta(\tau-t) d\tau = 1, t > 0$$

not define. $t=0$